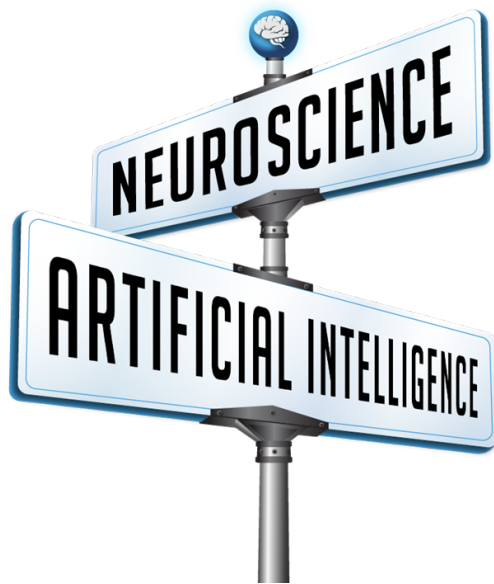


MODULE 1

Neuroscience as a Model for AI

Why the Brain Might Have Something to Teach Silicon



ITAI 4374 – Neuroscience and AI

Houston City College

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How to Use This Booklet

Welcome! This booklet is your **textbook chapter** for Module 1. Since this course has no physical textbook and no live lectures, this document contains everything you need to understand the material—complete explanations, visual cues, reflection questions, and connections to what you already know.

Think of this as a guided conversation, not a wall of text to memorize. The goal isn't to cram facts about neurons into your head. Instead, you'll build intuition for why studying the brain might help us design better AI systems.

What You Already Bring

You're a senior in the AI & Robotics program. You've built neural networks, worked with transformers, deployed agents, and probably debugged more Python than you'd like to remember. This module uses that background as your anchor—we'll constantly connect new neuroscience concepts to AI ideas you already understand.

Look for These Throughout the Booklet:

 **AI Parallel:** Connects neuroscience concepts to AI techniques you know

 **Common Misconception:** Clears up things that trip people up

 **We'll Return To This:** Flags topics covered in depth in later modules

 **Quick Check:** Pause-and-think questions (not graded—just for you)

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1. The Big Question: Why Study Brains to Build AI?

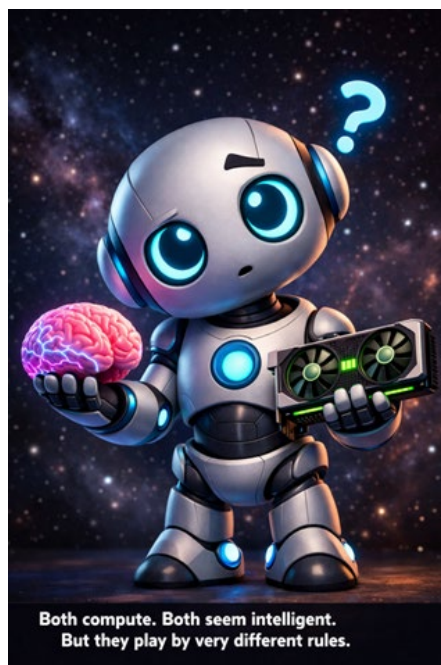


Figure 1.1: Both compute. Both seem intelligent. But they play by very different rules.

Let's start with an honest assessment of where we are. Modern AI is **incredible**. In the span of a few years, we've gone from 'chatbots that can barely hold a conversation' to 'systems that can write code, generate images, and pass professional exams.' You've seen this firsthand—you've built these systems.

And yet... something's off.

These same systems consume enough electricity to power small towns. They require months of training on curated datasets. They can confidently hallucinate facts. And when the world changes—a new event, a new context, a slightly different phrasing—they can fail in ways that surprise even their creators.

Meanwhile, sitting between your ears is a system that learned to recognize faces, understand language, navigate social dynamics, and plan for the future—all while running on about **20 watts**. That's less power than the laptop you're reading this on.

This course starts from a simple observation: general intelligence already exists in nature. The brain is proof that intelligence can be energy-efficient, adaptive, and robust. Instead of asking 'How do we make transformers bigger?', we're asking a different question:

What can we learn about intelligence by studying how the brain actually works?

AI Parallel

You've probably noticed that AI research tends to follow 'successful' architectures. Once transformers dominated NLP, everyone built transformers. This module asks: what alternatives might we be missing?

Quick Check *(pause and think—not graded!)*

- Think of an AI system you've worked with. What surprised you about its failures?
- If you had unlimited compute, would that solve all of AI's problems? Why or why not?

2. Intelligence: It's Not Just About Benchmarks

In AI, we love metrics. Accuracy. F1 scores. BLEU scores. Perplexity. These numbers give us something concrete to optimize. But here's the uncomfortable truth: a model that scores 99% on ImageNet can still be fooled by a sticker.

From a biological perspective, intelligence isn't about acing tests—it's about **functioning in the world over time**. That means:

- Learning from limited, messy, unlabeled experience
- Adapting when conditions change
- Integrating information across senses (vision + sound + touch + smell)
- Making decisions when you're uncertain
- Operating continuously—no 'deployment phase' where you're frozen

The brain doesn't encounter isolated tasks with clear start and end points. Life isn't a benchmark. It's a continuous stream of experience that requires real-time adaptation.

This distinction—between task-based performance and ongoing adaptive behavior—will be a theme throughout this course.

Quick Check *(pause and think—not graded!)*

- What's something humans do easily that even state-of-the-art AI struggles with?
- Why might optimizing for benchmarks lead to systems that fail in deployment?

3. The Hidden Assumptions in Modern AI



Figure 3.1: The rulebook nobody handed you—but everyone follows

Every engineering system is built on assumptions. Most of the time, we don't think about them—they're just 'how things are done.' But assumptions aren't laws of nature. They're choices.

Here are the assumptions baked into most modern AI:

Assumption #1: Intelligence = Discrete Steps

Neural networks process information in clearly defined stages. Data flows through layers. Transformers process tokens one attention step at a time. Everything is broken into neat, sequential operations.

The brain? It's a **continuous, parallel, asynchronous mess**. Neurons fire at different times. Signals loop back. There's no clear 'layer 1' and 'layer 2.' Processing and feedback happen simultaneously.

Assumption #2: Learning and Use Are Separate

The standard AI pipeline: collect data → train model → deploy model. Once deployed, the model is frozen. It doesn't learn from new experiences unless you explicitly retrain it.

Your brain? It's learning right now, as you read this sentence. There's no 'deployment mode' where you stop adapting.

Assumption #3: Dense Computation Is Normal

When you run inference on a neural network, every neuron in every layer activates. Even if you're just classifying a picture of a cat, billions of parameters are involved in the computation.

In the brain, only a tiny fraction of neurons fire at any moment. Most are silent, waiting for something relevant to happen. This is called sparse computation—and it's a big deal.

Common Misconception

"AI neurons are basically brain neurons, just simplified." Nope. Artificial neurons are mathematical functions optimized for GPUs. Biological neurons are living cells with electrical spikes, chemical signals, and thousands of synaptic connections that change over time. Inspired by $\neq$ equivalent to.

Quick Check *(pause and think—not graded!)*

- Which of these three assumptions feels most limiting to you? Why?
- Can you think of an application where continuous learning would be essential?

4. When Success Becomes a Straitjacket

Here's a paradox: the better something works, the less we question it.

Transformers have been so successful that they've become the default. Need to process text? Transformer. Images? Vision transformer. Audio? Audio transformer. Proteins? AlphaFold is built on transformers.

This success-driven convergence has benefits. Shared architectures mean shared tools, shared knowledge, and faster progress. But it also has a cost: we stop exploring alternatives.

When everyone's building transformers, research on **fundamentally different approaches** gets labeled as 'interesting but impractical.' Funding dries up. Talent moves elsewhere. The design space narrows.

Neuroscience offers a way out of this trap. The brain is proof that intelligence can be organized in a completely different way—one that didn't evolve to optimize benchmarks or exploit abundant compute, but to survive and adapt under severe resource constraints.

Studying the brain doesn't mean abandoning what works. It means expanding our sense of what's possible.

AI Parallel

Remember when RNNs were 'the' architecture for sequences? Then attention mechanisms changed everything. Major breakthroughs often come from questioning assumptions that everyone else accepted.

5. The 20-Watt Miracle: Energy as a Design Constraint

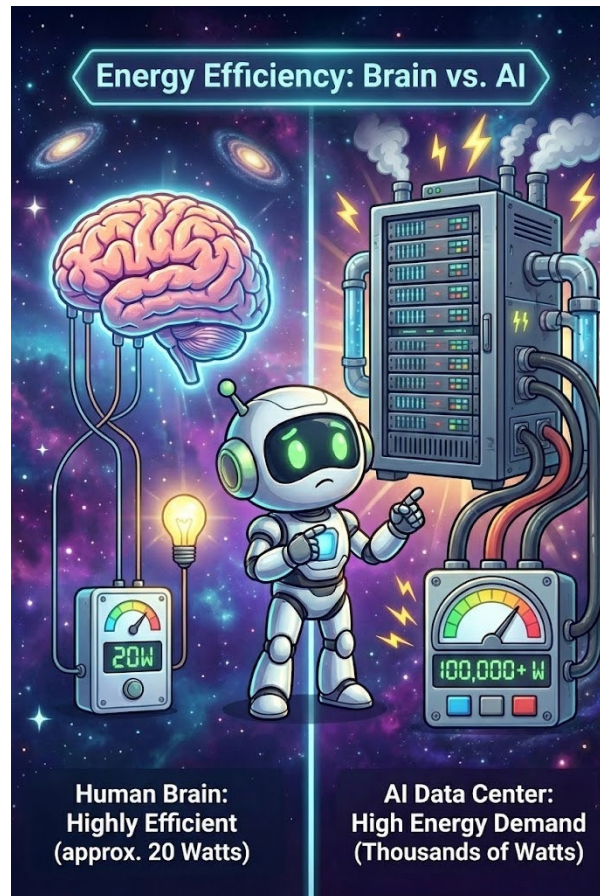


Figure 5.1: Same function, wildly different power bills

Let's talk numbers.

Training GPT-4 reportedly consumed enough energy to power thousands of homes for a year. Running inference on large models requires specialized hardware drawing hundreds of watts. Meanwhile, your brain runs on about **20 watts**—and it handles perception, memory, motor control, language, social reasoning, and creative thinking simultaneously.

This isn't just a fun fact. Energy scarcity fundamentally shapes how biological intelligence is organized.

When energy is limited, you can't afford to process everything all the time. You have to be selective. You have to prioritize. You have to turn off the parts you're not using. These constraints forced the brain to develop elegant solutions that we're only beginning to understand.

 Common Misconception

"Energy efficiency is just an engineering optimization—nice to have, not essential." In biological systems, energy constraints aren't afterthoughts. They shape the fundamental architecture of intelligence. Ignore them, and you end up with systems that work in data centers but fail in robots, edge devices, and the real world.

Any theory of intelligence that ignores energy is incomplete. And any AI system designed for the real world—not just the cloud—will eventually need to take energy seriously.

 Quick Check *(pause and think—not graded!)*

- If you had to run AI on a device with a 10W power budget, what would you sacrifice first?
- Why do you think evolution prioritized energy efficiency over raw computational power?

6. Two Ways to Compute: Dense vs. Sparse

Now we get to one of the most important distinctions in this course.

Dense Computation (The AI Way)

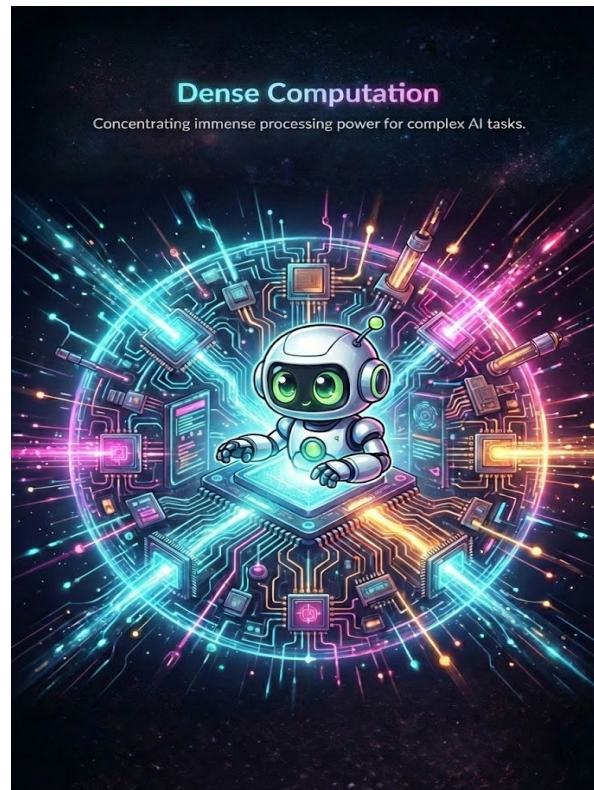


Figure 6.1: Dense computation—everyone participates, whether they're relevant or not

In standard neural networks, every forward pass activates most or all of the neurons. Processing an image? Every layer fires. Generating a token? Attention spans the entire context window.

This design makes sense for GPUs, which are optimized for massive parallel operations. But it means energy usage is constant and high, regardless of how 'easy' the input is.

AI Parallel

This is why techniques like pruning, quantization, and Mixture of Experts (MoE) are hot topics. They're attempts to add sparsity to dense architectures—with varying success.

Sparse Computation (The Brain Way)

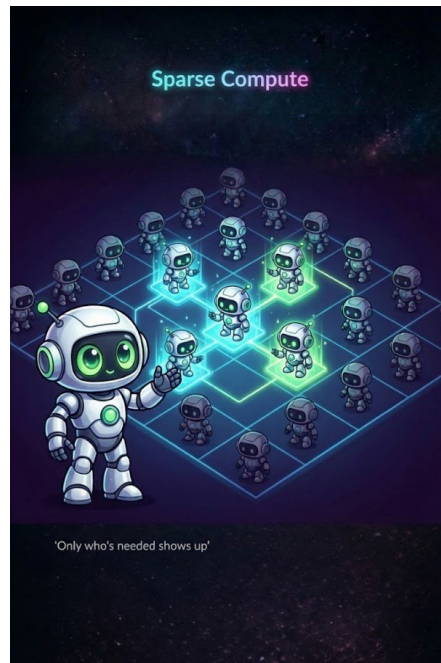


Figure 6.2: Sparse computation—selective activation saves energy and focuses resources

In your brain, only about **1-10% of neurons** fire at any given moment. The rest are silent. Neurons only activate when specific conditions are met—when something relevant happens that requires their contribution.

This isn't a bug; it's a feature. Sparse computation allows the brain to:

- Conserve energy (silent neurons don't cost much)
- Scale efficiently (more neurons doesn't mean more energy)
- Focus processing on what matters
- Support different tasks without interference

Why This Matters

The dense vs. sparse distinction isn't just about efficiency. It reflects different **philosophies of intelligence**. Dense systems assume that processing everything is valuable. Sparse systems assume that being selective is the key to intelligence.

Neuroscience suggests that intelligent behavior depends more on what you process than on processing everything.

Quick Check (pause and think—not graded!)

- Why might sparse computation be essential for real-time robotics?
- What AI techniques have you used that incorporate some form of sparsity?

7. A Brief History: How AI and Neuroscience Grew Apart



Figure 7.1: A relationship history—together, apart, and... maybe back together?

This wasn't always a course about 'reconnecting' AI and neuroscience. In the early days, they were the same field.

The Beginning: We Were One (1940s-1960s)

The perceptron, one of the first AI models, was explicitly designed to mimic biological neurons. Researchers like McCulloch, Pitts, and Rosenblatt drew directly from neuroscience. The goal was to understand and recreate how the brain thinks.

The Divergence: Math Takes Over (1970s-2000s)

As AI matured, a split emerged. Progress came from mathematical optimization, not biological accuracy. Backpropagation enabled training deep networks—but it's not how brains learn. The focus shifted from 'how does biology do it?' to 'what math makes the loss go down?'

This shift made sense. Mathematical approaches were easier to analyze, implement, and scale on digital hardware. Biological realism was seen as an unnecessary constraint.

The Present: Biology Makes a Comeback

Today, as AI systems hit walls—energy costs, adaptation limits, robustness problems—researchers are looking back at biology. Not to copy neurons exactly, but to ask: **What design principles did evolution discover that we're missing?**

AI Parallel

Spiking Neural Networks (SNNs), neuromorphic chips (Intel's Loihi, IBM's TrueNorth), and biologically-inspired learning rules are all examples of AI reconnecting with neuroscience. We'll explore some of these in later modules.

8. Transformers: A Success Story (With Footnotes)

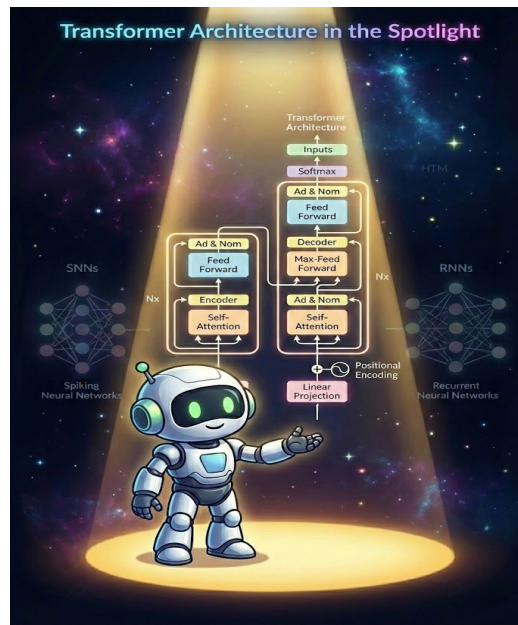


Figure 8.1: The star of the show—but not the only act

You know transformers well. Self-attention. Positional encoding. Feed-forward layers. Scaling laws. You've probably built applications on top of transformer-based models.

Transformers deserve their success. They demonstrate that intelligence-like behavior can emerge from scale, dense computation, and massive datasets. They've enabled breakthroughs in NLP, vision, code generation, and more.

But let's read the footnotes:

Footnote 1: Transformers treat intelligence as sequence prediction. Is that the only way?

Footnote 2: Learning happens offline. Deployed models don't learn from their experiences.

Footnote 3: Memory is stored in weights or retrieval systems. Is that how biological memory works?

Footnote 4: Dense attention scales quadratically. There's a reason people keep inventing 'efficient transformers.'

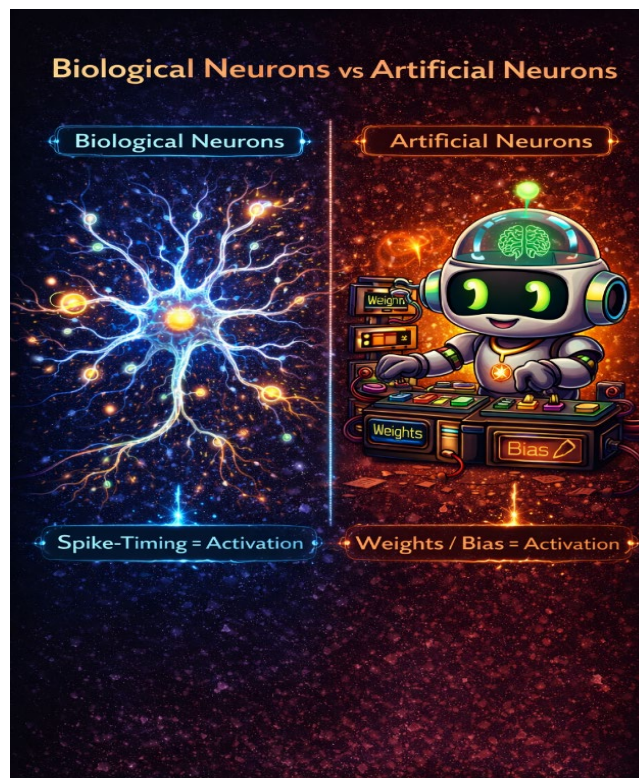
The success of transformers doesn't mean these assumptions are universal. It means they work well in certain conditions—conditions we've engineered (massive compute, curated data, controlled deployment).

What about conditions we haven't engineered? Edge devices. Robots in the field. Systems that need to learn for years without retraining.

 **Quick Check** (pause and think—not graded!)

- What's one assumption in transformer design that you've always wondered about?
- Can you think of an application where transformer assumptions would break down?

9. Learning FROM the Brain, Not Copying It



Let's be clear about what this course is not:

- We're not trying to build an artificial brain
- We're not claiming biology is 'better' than engineering
- We're not abandoning everything that works in modern AI

Instead, we're using neuroscience as a source of constraints and design principles.

Constraints inspire creativity. When evolution had to build intelligence on 20 watts, it discovered solutions we might never think of. When neurons could only communicate through sparse spikes, it led to efficient, event-driven processing. When memory had to be stable yet flexible, it led to systems that consolidate and reconsolidate information.

By studying these constraints, we expand the space of possible AI designs rather than narrowing it.

Key principles from neuroscience that we'll explore:

Energy efficiency: Operating under strict power budgets

Lifelong learning: Continuous adaptation without catastrophic forgetting

Robustness to noise: Reliable performance despite noisy, incomplete information

Sparse, selective processing: Computing only what matters

Integrated memory: Memory that shapes perception and action in real-time

10. Memory: The Core of Intelligence (Sneak Preview)



Figure 10.1: Two philosophies of memory—frozen storage vs. active reconstruction

Any intelligent system must retain information over time. But how?

Memory in AI (How You've Learned It)

In AI systems, memory typically means: weights (learned during training and frozen), embeddings (stored representations), external databases (for RAG), or context windows (limited and session-based). These forms of memory are usually **static**—they don't change during normal operation.

Memory in Biology (Spoiler Alert)

In the brain, memory is dynamic, contextual, and deeply integrated with everything else. Memories aren't retrieved from storage—they're reconstructed. Recall changes memories. Sleep consolidates them. Context shapes what you remember.

We'll Return To This

Modules 6-7 go deep on memory systems and learning mechanisms: synaptic plasticity, memory consolidation, the hippocampus, and more. For now, just note that memory is central to intelligence—not an afterthought.

Common Misconception

"Brains don't use data or examples." Humans absolutely learn from experience! The difference isn't the absence of data—it's the absence of curated datasets, explicit labels, and clean training/deployment separation. Life doesn't hand you MNIST.

11. Learning Without a 'Training Phase'

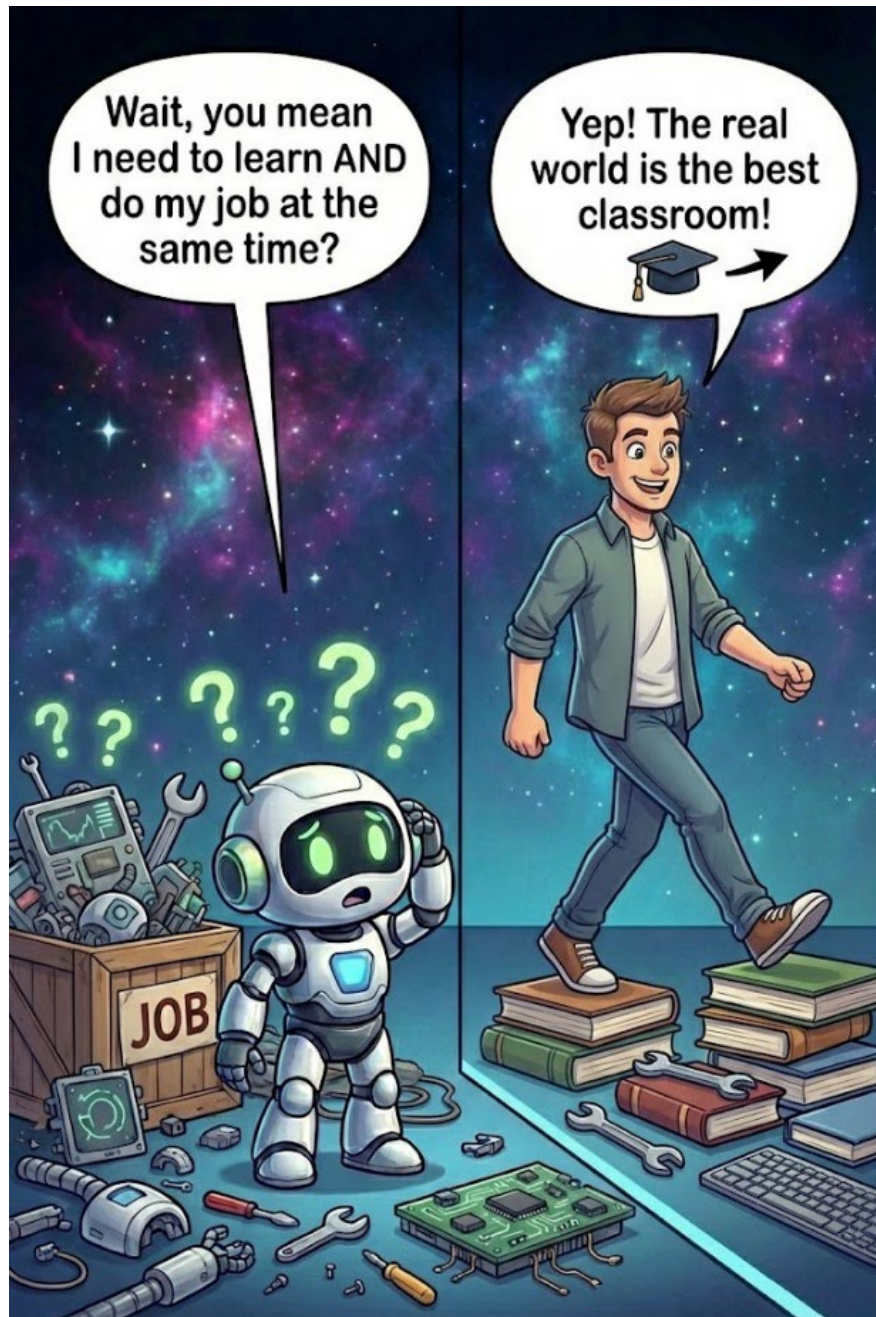


Figure 11.1: The biological approach—learning happens during operation, not before

Think about how you learned to ride a bike. Did someone give you a dataset of 10,000 bike-riding examples? Did you train in a simulation for weeks before your first ride? Did you freeze your skills after training and deploy yourself?

Of course not. You got on the bike, fell, adjusted, fell less, adjusted more, and eventually learned—while doing the thing, not before.

The AI Pipeline

Standard AI: Collect data → Preprocess → Train for hours/days/weeks → Evaluate → Deploy (frozen) → Retrain later if needed

This pipeline assumes a **clean separation** between learning and use. But for the brain, there is no separation.

The Biological Approach

Learning happens continuously during operation. You're learning right now, as you read. You'll remember some of this tomorrow, and that memory will influence how you think about AI. There's no 'training mode' where you stop functioning and just absorb data.

This continuous, embedded learning is one of the brain's superpowers—and one of the hardest things for AI to replicate.

Common Misconception

"Learning only happens during training." True for most AI systems, but not for brains. Biological learning is lifelong, contextual, and interleaved with action. This is why 'continual learning' is such a hot research topic.

We'll Return To This

Module 7 focuses specifically on learning: synaptic plasticity, Hebbian learning, error-driven learning, and how brains balance stability with adaptation.

12. "But What About...?" — Modern AI Developments

By now, you might be thinking: *"Wait—haven't recent AI developments already addressed some of these limitations?"* If so, good. That's exactly the kind of critical thinking this course demands.

Let's directly address the cutting-edge developments you're probably wondering about: reasoning models, agentic systems, RAG, and chatbot memory. Understanding what these innovations actually do—and what they don't—will sharpen your understanding of why neuroscience still matters.

Reasoning Models: o1, o3, and Chain-of-Thought

OpenAI's o1 and o3 models, along with chain-of-thought (CoT) prompting techniques, have demonstrated impressive improvements in complex reasoning tasks. These models appear to *"think step by step"* before answering.

So doesn't this address the "discrete steps" assumption?

Not quite. Here's what's actually happening:

- The model generates explicit reasoning tokens as part of its output sequence
- This is still discrete, sequential token generation—just more of it
- The "thinking" is serialized into text, not parallel processing
- There's no continuous, asynchronous processing happening simultaneously

The brain's reasoning involves **parallel activation across multiple regions simultaneously**, feedback loops that operate in real-time, and integration of emotional, sensory, and memory systems. When you solve a problem, you're not generating a linear chain of internal "tokens"—your prefrontal cortex, hippocampus, and other regions are engaged in a dynamic, recurrent dance.

We'll Return To This

Module 4 explores how biological neural circuits implement reasoning through recurrent connections and parallel processing—mechanisms fundamentally different from autoregressive token generation.

Agentic AI: Continuous Learning at Last?

Agentic AI systems—autonomous agents that can plan, use tools, and pursue goals over extended interactions—represent one of the most exciting developments in AI. You've probably built agents using LangChain, AutoGPT, or similar frameworks.

Don't agents learn continuously as they interact with their environment?

Let's be precise about the mechanism:

- Agents accumulate information within a session (context window)
- They can write to external memory stores (databases, files)
- They adapt their behavior based on feedback within an episode

But the underlying model's weights don't change. When the agent encounters a novel situation, it's not modifying its neural network. It's retrieving information and applying in-context learning—powerful, but fundamentally different from biological learning.

Consider what happens in your brain when you learn something new:

- Synaptic connections physically strengthen or weaken
- New neural pathways form
- Memory consolidation occurs during sleep
- The learning is integrated into your existing knowledge structure

Agents simulate continuity through *external scaffolding* (memory stores, tool use, planning modules). The brain achieves continuity through *intrinsic plasticity*—the architecture itself changes.

AI Parallel

Online learning and continual learning are active research areas attempting to update model weights during deployment. But they face the catastrophic forgetting problem: new learning overwrites old knowledge. The brain solves this elegantly through complementary learning systems—something we'll explore in Module 7.

RAG: External Memory, Not Biological Memory

Retrieval-Augmented Generation (RAG) has become a standard technique for giving language models access to external knowledge. Instead of relying solely on parameters, the model retrieves relevant documents and incorporates them into its response.

Isn't RAG basically giving AI a form of memory?

RAG is genuinely useful, but it's architecturally very different from biological memory:

RAG Memory	Biological Memory
Explicit retrieval from database	Reconstruction from distributed patterns
Storage separate from processing	Memory integrated with perception/action
Retrieved content unchanged	Recall modifies the memory itself
Semantic search via embeddings	Content-addressable, context-dependent
No consolidation or forgetting	Active consolidation, selective forgetting

The most fundamental difference: in RAG, **the act of remembering doesn't change what's remembered**. In biological memory, every recall is a reconstruction that can modify the memory trace. This isn't a bug—it's a feature that enables generalization, updating, and integration of new information.

We'll Return To This

Module 6 covers memory systems in depth: the hippocampus, memory consolidation during sleep, the difference between episodic and semantic memory, and why "perfect" memory might actually be a disadvantage.

Chatbot Memory: Persistence ≠ Learning

Modern chatbots like Claude, ChatGPT, and others now offer "memory" features that persist information across conversations. You can tell them your name, preferences, or ongoing projects, and they'll remember in future sessions.

Doesn't this mean AI is learning after training?

Let's be precise about the mechanism:

- User information is stored in an external database or memory system
- This stored information is retrieved and injected into the context window

- The model processes this information as if it were part of the conversation
- The base model's weights remain completely unchanged

This is *persistence*, not *learning*. It's more analogous to you writing notes in a notebook and reading them later—useful, but not the same as forming new neural pathways.

When you learn your friend's name, you don't write it in an external database and look it up each time you see them. The information becomes part of your neural structure, automatically activated by relevant contexts, integrated with your knowledge of that person.

⚠ Common Misconception

"The AI remembered my birthday—it must be learning!" What's actually happening: your birthday was stored in a database, retrieved at conversation start, and inserted into context. The model processed it like any other text. No synapses were modified. No learning occurred.

The Deeper Point: Scaffolding vs. Architecture

All of these innovations—reasoning models, agents, RAG, chatbot memory—share a common pattern: they add capabilities through external scaffolding rather than changing the fundamental architecture.

This is not a criticism. Scaffolding works. It enables remarkable capabilities. But it's important to understand the difference:

Scaffolding Approach (Current AI)	Architectural Approach (Biology)
Static core + external add-ons	Plastic core that changes with experience
Complexity grows with capabilities	Same architecture scales to new capabilities
Multiple systems to coordinate	Unified system with emergent coordination
Brittleness at integration points	Graceful degradation

The brain doesn't need a separate "memory module," a "reasoning module," and a "tool use module" that communicate through APIs. These capabilities emerge from the same underlying substrate—neurons, synapses, and the dynamic patterns of activity they support.

This is why neuroscience still matters even as AI capabilities improve. We're not asking "*Can AI do X?*" but rather "*What's the most elegant architecture for achieving X?*" Current approaches work, but they may not be the only way—or the best way for all applications.

 Quick Check (pause and think—not graded!)

- What's the key difference between a chatbot's "memory" and your memory of a friend?
- Why might scaffolding approaches struggle in resource-constrained environments (like robots)?
- Can you think of a capability that would require architectural changes, not just better scaffolding?

13. What Module 1 Has Established

Let's step back and see where we are.

This module wasn't about teaching you neuroscience facts. It was about shifting perspective. Here's what we've established:

- 1. Intelligence is constrained.** Energy, time, noise, and biological limits shape how intelligence can be organized. These aren't obstacles to work around—they're design parameters.
- 2. Modern AI reflects specific assumptions.** Discrete computation, separate training and deployment, dense processing, static memory. These work well, but they're not laws of nature.
- 3. The brain offers a different example.** Not a blueprint to copy, but proof that intelligence can work in fundamentally different ways.
- 4. Neuroscience expands the design space.** By studying biological constraints, we discover alternatives we wouldn't find by just scaling up what works.
- 5. Modern innovations address symptoms, not architecture.** Reasoning models, agents, RAG, and chatbot memory add capabilities through scaffolding—powerful, but different from how biology achieves these capabilities.

This foundation sets up everything that follows. Starting in Module 2, we'll explore how intelligence is actually organized in the brain—the regions, the circuits, the principles.

14. Looking Ahead to Module 2

Module 1 answered: Why should AI practitioners care about neuroscience?

Module 2 will answer: **How is intelligence organized in the brain?**

You'll be introduced to:

- Major brain regions and what they do
- How different parts work together (hint: not like a typical software stack)
- Principles of neural organization that might inspire AI design
- Why 'one function per region' is way too simple

We'll focus on organization—the architecture—not the low-level biology. You don't need to memorize neurotransmitters. You do need to understand how different processing systems interact.

See you in Module 2!



15. Glossary Lite (Module 1)

Quick-reference definitions for terms introduced in this module.

Computation

The process of transforming input information into output. In AI: numerical operations on hardware. In brains: patterns of neural activity. Don't confuse 'computation' with 'math equations'—it's about information processing.

Constraint

A limitation that shapes what a system can do and how it must do it. Examples: energy budget, time limits, noise tolerance. Constraints aren't just problems—they drive design innovation.

Dense Computation

A style where most processing units are active most of the time. Standard in modern neural networks. Efficient on GPUs but energy-hungry.

Sparse Computation

A style where only a small fraction of units are active at any moment. The brain's default mode. Conserves energy and focuses resources.

Static Memory

Information that stays fixed during normal operation. In AI: trained weights that don't change during inference.

Dynamic Memory

Information that changes with experience and use. The brain's approach—memories are reconstructed, not just retrieved.

Training vs. Deployment

The separation in AI systems between learning (training) and use (deployment). Brains don't have this separation—learning is continuous.

Scaffolding (in AI)

External systems added to a static model to provide additional capabilities (memory stores, tool use, planning modules). Contrast with intrinsic architectural capabilities.

Catastrophic Forgetting

The tendency of neural networks to completely forget previously learned information when learning new information. A major challenge in continual learning.

In-Context Learning

The ability of large language models to adapt their behavior based on examples or instructions provided in the prompt, without changing model weights.

16. Closing Reflection

Before you close this booklet, take a few minutes with these questions. They're not graded—they're for you. Write your thoughts in a notebook or just think them through.

This is the kind of thinking that separates people who understand AI from people who just use it.

 **Quick Check** (*pause and think—not graded!*)

- Which assumption in modern AI feels the least inevitable to you now?
- If you had to design an AI system with a 10W power budget, what would you prioritize?
- Where do you think neuroscience will matter most as AI leaves the lab and enters the real world?
- What's one thing from this module that changed how you think about 'intelligence'?
- How would you explain the difference between 'scaffolding' and 'architecture' to a friend?

And finally, a thought to carry forward:

If building intelligence were just about stacking more layers and GPUs, evolution could have saved itself a few million years.

If intelligence is constrained by energy, memory, and adaptation—and if different systems satisfy those constraints in different ways—then no single architecture should dominate our understanding of intelligence.

This course invites you to explore intelligence beyond today's defaults.

— End of Module 1 —

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